



Threshold concepts for writing with AI: Experimentation, expertise, agency[☆]

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ABSTRACT

How does writing with generative AI transform our experiences as thinkers, learners, and communicators? This study investigates the question through threshold concepts, transformative ideas that prove troublesome yet shape disciplinary participation, previously identified in writing studies through retrospective analysis. We propose that technological disruption creates conditions for observing new concepts as they emerge within communities of practice. To test this approach, we designed an experimental “AI and Writing” course. Across two iterations (Fall 2023 and Fall 2024), 38 undergraduate students from 22 majors engaged in scaffolded challenges and self-directed projects, developing strategic approaches to AI-assisted composing. Analysis of students’ final reflections, using Meyer and Land’s characteristics as sensitizing criteria, revealed three concepts essential for productive AI engagement: writing with AI is an experimental process; writing with AI requires expertise and dialogue; and writing with AI should augment rhetorical agency. These transformations prove troublesome because they contradict expectations shaped by prior experience: better tools should require less effort, linguistic fluency signals substantive knowledge, and assistance either wholly helps or harms learning and development. The findings show how AI transforms perennial questions in writing studies and composition, and they offer insights for navigating the threshold between productive collaboration and problematic dependency.

As with previous eras of technological change, the associated disciplines of writing studies and composition find themselves on the frontier of exploring how generative AI technologies will transform literate practice. Our educational mission and courses serve as gateways through which students gain academic and professional literacies that AI now promises to augment or erode. Since ChatGPT’s release in November 2022, our field has responded swiftly with two special issues in *Computers and Composition* dedicated to “Composing with generative AI” in 2024 and “Digital multimodal composing in the era of artificial intelligence” in 2025, framing this emerging conversation.

In their introduction to the first special issue, [Ranade and Eyman \(2024\)](#) foregrounded how human-centered applications of AI promise to provide personalized, on-demand feedback and assistance that can have positive impacts on the productivity, quality, and creativity of student writing. The challenge is understanding how we can use these tools to productively support “writing as communication, writing as learning, and writing as thinking” (p. 3). However, Ranade and Eyman also raised serious concerns about

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how access to these tools will impact student self-efficacy, motivation, and creativity. This raises the question of whether AI assistance will ultimately empower or undermine the development and exercise of writing skills.

Complementing this work, [Lim et al. \(2025\)](#) introduced the second special issue by positioning generative AI within the tradition of digital multimodal composition, where prior research has demonstrated positive effects on students' literacy development even as new technologies disrupt established practices. On this point, [Lim et al. \(2025\)](#) invoked Bolter and Grusin's concept of remediation, describing the process through which digital media tools enable users to "integrate, transform, and revitalize previous media formats" (p. 2). Writers leverage existing composing skills to engage new media, adapting what they know to unfamiliar contexts, while new media simultaneously transform literate practices, revealing new possibilities and constraints that reshape how writers understand composing techniques, media, and genres. This perspective calls for work exploring how generative AI may both extend and challenge our understanding of writing processes and skills.

Thus, early scholarship on generative AI allows us to frame a compelling research question: how does writing with AI remediate our relationships to ourselves as thinkers, learners, and communicators? This question marks a liminal threshold requiring active sense-making within disciplinary knowledge and practice. Threshold concepts, articulated for writing studies in *Naming What We Know* ([Adler-Kassner & Wardle, 2015](#)) and *(Re)Considering What We Know* ([Wardle & Adler-Kassner, 2020](#)), provide such a framework. Threshold concepts are transformative ideas that prove troublesome to grasp yet transformative once understood, enabling learners to develop "knowledge-making practices and ways of seeing in a discipline through expertise" ([Adler-Kassner & Wardle, 2020](#), p. 16). In an era in which generative AI seems poised to transform perennial questions in writing studies and composition, re-examining threshold concepts offers a way to rediscover writing studies' foundational principles, and to identify the conceptual shifts that distinguish productive use from problematic dependency.

This study investigates threshold concepts for AI and writing by observing how students develop literate practices through experiential learning. We designed an "AI and Writing" course as a site of shared experimentation at the intersection of composition pedagogy and generative AI technologies, treating it as a community of practice in which sustained engagement and peer discussion provided conditions for new practices and understandings to emerge together. Students worked through scaffolded AI literacy activities and self-directed creative projects, learning to engage AI tools as strategic resources for multimodal composing. Our analysis examines students' final course reflections, attending to moments when students recognized their thinking had changed and could articulate what had shifted. These patterns of conceptual transformation can help other scholars and learners cross similar thresholds.

Threshold concepts and writing studies

Threshold concepts were developed in [Meyer and Land's \(2005\)](#) investigation into why certain ideas prove difficult for learners to grasp, yet once understood, fundamentally transform their ability to participate in disciplinary thinking. These concepts function as "conceptual gateways" or "portals" that lead to previously inaccessible ways of understanding subject matter. [Adler-Kassner and Wardle \(2020\)](#) have described them as "lenses learners must see through and see with to be successful" within communities of practice (p. 3). [Meyer and Land \(2005\)](#) elaborated five characteristics distinguishing threshold concepts from routine learning: they are *troublesome* because they challenge and conflict with existing knowledge; *bounded* within particular disciplinary communities; *integrative* in exposing previously unrecognized connections among phenomena; *transformative* in reshaping not merely what learners know but how they engage disciplinary discourse and practice; and, *irreversible* in that, once understood, they resist being "unseen" (pp. 373–374).

Writing studies embraced threshold concepts through a collaborative effort that became *Naming What We Know* ([Adler-Kassner & Wardle, 2015](#)). Through deliberation among writing teacher-researchers, the project identified five overarching threshold concepts and thirty-seven supporting concepts. *(Re)Considering What We Know* later examined how these concepts circulated across educational contexts, establishing them not as fixed doctrine but as generative frameworks for ongoing conversations about expertise in writing ([Adler-Kassner & Wardle, 2020](#)). [Table 1](#) presents the original five major threshold concepts from *Naming What We Know* which prove crucial for understanding AI's disruption of writing practice.

Troublesomeness and liminality are the foundational characteristics of threshold concepts. Threshold concepts are not synonymous with the core concepts or key ideas of a discipline; rather, they register as points of disruption and transformation in the learning process. As [Adler-Kassner et al. \(2012\)](#) have argued learners begin in a preliminary phase, by encountering disciplinary concepts that

Table 1
Five threshold concepts from naming what we know.

Concept	Description
TC 1: Writing Is a Social and Rhetorical Activity	Writing is relational and responsive; meaning emerges through engagement with audiences, purposes, and contexts
TC 2: Writing Speaks to Situations through Recognizable Forms	Genres emerge from and respond to rhetorical situations; forms mediate between writers and contexts
TC 3: Writing Enacts and Creates Identities and Ideologies	Writers construct selves and values through discourse; composing within disciplines shapes who writers become as participants in those communities
TC 4: All Writers Have More to Learn	Development continues throughout writerly lives; adaptation, revision, and failure prove productive
TC 5: Writing Is (Also Always) a Cognitive Activity	Composition involves managing memory, embodied cognition, and metacognition

Note. Adapted from *Naming What We Know: Threshold Concepts of Writing Studies* ([Adler-Kassner & Wardle, 2015](#)).

challenge prior knowledge, beliefs, and practices. The process is as much about existing tacit knowledge becoming available for examination and re-articulation as it is about acquiring new knowledge and skills. Through liminal phases, the learner begins to work with the new knowledge and skills more directly and to discover how to enact them; this process is characterized by complex, recursive, and uneven progress. In the postliminal phase, the learner develops greater “metacognitive awareness” which informs epistemic understanding and learning transfer, allowing them to abstract the knowledge or skill and apply it across diverse contexts. Threshold concepts have a characteristically diagnostic value, directing our attention to those learning challenges that most determine whether students gain access to disciplinary ways of seeing, thinking, and acting.

In writing studies, threshold concepts have proven generative because they have helped scholars and educators attend to critical sites of learning across the contexts of teacher preparation, curriculum design, and instructional delivery. However, as [Adler-Kassner and Wardle \(2020\)](#) have acknowledged, our existing writing studies threshold concepts represent “the most frequently agreed-upon knowledge of a field at a particular moment in time,” identifying widely accepted ideas through retrospective analysis (p. 24). This approach is inherently slow moving and cannot readily keep pace with rapidly evolving phenomena like generative AI. Yet, if threshold concepts are identifiable through the friction they generate between disciplinary understanding and prior knowledge, then observing that conflict as it emerges within learning communities offers a useful way to study technological change.

Communities of practice offer a valuable site for this kind of real-time observation. In writing studies, [Adler-Kassner and Wardle \(2020\)](#) have argued for using the term “communities of practice” instead of “disciplines,” highlighting that writing threshold concepts function “in sites well beyond academic disciplines” and particularly at “specific intersections among distinct communities of practice” where different disciplinary expertise and knowledge structures meet (pp. 3–4). This argument reflects the broad applicability of writing expertise across disciplines and professions, and the practical reality that most writing programs and courses serve students from diverse degree programs. Communities of practice are also ideal for this inquiry because of the model of learning they describe. [Wenger \(1998\)](#) defined communities of practice as groups united by what he called “mutual engagement” in a shared enterprise, through which participants develop “a shared repertoire of resources,” including strategies, vocabulary, and practical knowledge (p. 73). Learning is emergent through social participation rather than primarily through expert-to-novice transmission. During periods of technological change, this model can inform pedagogical approaches emphasizing shared inquiry, where educators and students work together to experiment, make sense of disruptions, and develop new knowledge and practices.

Threshold concepts provide the diagnostic lens for the current study, while communities of practice describe the conditions under which threshold learning becomes observable. As diagnostic instruments, threshold concepts map the liminal journey from conceptual conflict to metacognitive reorientation, identifying where prior knowledge most strongly conflicts with emerging disciplinary understanding and where successful reorientation produces the broadest consequences for how learners participate in a field. Communities of practice, in turn, create contexts for students to engage in active sense-making and social comparison through which tacit assumptions become explicit, nascent understandings get tested against peers’ experiences, and metacognitive awareness gradually develops. By treating our “AI and Writing” course as a community of practice and using [Meyer and Land’s \(2005\)](#) threshold concept characteristics as diagnostic criteria, we examine students’ final reflections for the characteristic signs of threshold concepts: moments when students articulate prior assumptions, describe the troublesome recognition that those assumptions require revision, and characterize the resulting shifts as ones they cannot imagine reversing. The emergent concepts our analysis identifies extend existing writing studies threshold concepts, documenting learners’ liminal journeys at the intersection of established disciplinary knowledge and unprecedented technological affordances.

Liminal thresholds for writing with AI

Emerging scholarship on AI-assisted composing demonstrates how generative AI is creating new troublesome conflicts with each of the five foundational threshold concepts established in *Naming What We Know*. Mapping that literature onto this framework allows us to anticipate where learners are most likely to encounter conceptual challenges. There are three areas where these conflicts are most concentrated: how writers manage cognitive demands and strategic oversight, how they develop rhetorical and disciplinary authority in relation to AI’s formal fluency, and how they exercise distributed agency in human-AI composing.

Writing with AI as cognitive activity

Cognitive capacity and working memory represent primary constraints on writing development; these are constraints that AI can both moderate and complicate. The threshold concept “Writing Is (Also Always) a Cognitive Activity” (TC 5) describes this foundational dynamic. Working memory is where writers hold fleeting information, juggle competing demands, and manage multiple tasks. As [Dryer et al. \(2015\)](#) observed, working memory appears fairly fixed in capacity and zero-sum in nature, creating predictable patterns: when writers encounter unfamiliar tasks, the additional cognitive load reduces performance in areas where they normally demonstrate competence (p. 73). Over time, writers develop automaticity as practiced routines recede from conscious attention, freeing working memory for higher-order composing goals (p. 74). Managing these dynamics requires metacognition: an awareness of one’s own cognitive processes that allows writers to assess which skills apply in novel situations, decide where to focus effort, and develop techniques for managing cognitive load (p. 76).

How do writers manage these constraints in practice? [Overstreet \(2022\)](#) argued that understanding writing as embodied cognition—distributed across physical actions, affective states, and technological interfaces—helps explain how technologies support writing as cognitive activity. Given working memory’s biological constraints, emerging technologies enable what Overstreet described as “increasingly sophisticated acts of cognitive offloading” (p. 8). Writers externalize mental work onto paper, screens, and software:

outlines hold emerging structure, drafts enable revision, and citation software track sources. These offloading strategies are fundamental to how writing works, enabling writers to handle rhetorical complexity that would otherwise exceed cognitive capacity.

LLMs extend this relationship in a distinct way: they can directly engage in rhetorical labor, not merely supplement memory storage. Knowles (2024) developed the concept of rhetorical load sharing (RLS) to describe how the cognitive and rhetorical demands of composing can be distributed between human and AI collaborators across the five canons of rhetoric (p. 4). He acknowledges the risk that writers seeking to offload cognitive effort might ask AI to handle the entire rhetorical load. He argues instead that ethically-grounded approaches require human collaborators to retain “the majority of the rhetorical load” in order to preserve rhetorical judgment and authorship (p. 8). Writers might therefore use AI to reduce routine cognitive labor, such as generating initial draft content from detailed human-created outlines, without surrendering meaningful control over invention or arrangement. This challenge is fundamentally metacognitive: writers must decide not only what to write but how to distribute labor between human and machine, and they must maintain strategic control over rhetorical choices even as they delegate aspects of execution.

Hence, productive AI use entails a counterintuitive trade-off: reduced cognitive effort in task execution demands increased metacognitive effort in planning and strategic oversight. Previous work has demonstrated what strategic oversight looks like in practice: writers must actively plan before delegating to AI, iterate through successive outputs, and evaluate results against their own rhetorical goals rather than accepting generated content uncritically (Anders & Dux Speltz, 2025). This threshold proves troublesome precisely because it contradicts intuitive expectations. Students may approach AI imagining technology as a path to easier writing, yet what productive engagement actually requires is a shift from passive offloading to active orchestration of distributed cognitive resources.

Writing with AI as social and rhetorical

Writing is relational, responsive, and involves participating in ongoing conversations with others (Roozen et al., 2015). The threshold concept “Writing Is a Social and Rhetorical Activity” (TC 1) names this foundational understanding. It is reflected in core disciplinary concepts such as the rhetorical triangle which foregrounds the dynamic relationships of writer, audience, and text within situated communicative contexts. Writing is more than sentence-level craft; it involves “mulling over a problem, thinking with others, and exploring new ideas or bringing disparate ideas together” (Roozen et al., 2015, p. 19). Writers must make their thoughts material in and through language, and audiences must devote attention and interpretive effort to reconstructing that meaning. Effective writing thus requires understanding the social and discourse conventions that ground communicative relationships.

Before writers choose words, they must identify situations and the nature of the communication they seek to make. In this work, they draw upon a closely related threshold concept, “Writing Speaks to Situations through Recognizable Forms” (TC 2) (Bazerman et al., 2015, p. 35). Genre recognition provides essential clues for this work: it helps writers and readers understand what kinds of messages a document may contain, what situation it addresses, and what roles and relations it establishes between participants. In writing studies, genres are understood not as static templates but as “the visible effects of human action, routinized to the point of habit in specific cultural conditions” (Bazerman et al., 2015, p. 39). The formal elements that characterize genres represent accumulated evidence of how writers have employed familiar moves in accordance with reader expectations, institutional norms, and disciplinary values. The relationship is reciprocal: disciplines shape the writing their members produce, and that writing in turn shapes disciplinary knowledge-making itself.

AI-assisted writing intensifies these longstanding challenges. When AI generates context-appropriate text, mimics disciplinary forms, or adapts tone for specified audiences, it can appear to handle social and rhetorical demands autonomously. This threshold proves troublesome because linguistic fluency creates powerful illusions of competence. Mahowald et al. (2024) have argued that LLMs excel at formal linguistic competence while lacking functional competence: the ability to reason about situations, verify claims, or model the world. Because humans tend to expect that facility with language signals facility with thought, this can lead us to “mistakenly attribute intelligence and intentionality” to these technologies (Mahowald et al., 2024, p. 518). When AI generates text that sounds authoritative and contextually appropriate, students may assume it reflects genuine understanding of disciplinary conventions, audience needs, and rhetorical situations. This conflation of fluency with knowledge and competence makes deferring to AI outputs feel reasonable rather than problematic.

Recognizing the importance of situated, disciplinary knowledge is crucial for empowering students to become discerning users of AI. Bearman and Ajjawi (2023) have argued that working with AI’s “black box” requires developing and exercising “evaluative judgement” to assess “whether work is meeting quality standards” (p. 9). Writers cannot simply verify facts; they must make discerning judgements about whether outputs meet the disciplinary frameworks defining legitimate knowledge within communities of practice. Evaluative judgment in this sense goes beyond generic fact-checking toward what Bearman and Ajjawi (2023) described as the importance of developing discipline-specific understanding of how success is constituted and what makes communication effective and legitimate within particular social contexts.

This liminal threshold requires recognizing that effective collaboration is grounded in human expertise. Rather than deferring to apparent technological omniscience, writers must assume rhetorical and intellectual authority, actively directing AI through their developing expertise in audience awareness, genre knowledge, and disciplinary conventions. Writers must supply what AI fundamentally cannot access: situated judgment about what makes communication effective and legitimate within particular rhetorical situations and communities of practice. From this grounding in situated expertise, ethical use becomes a matter of ongoing dialogue rather than fixed rules. As Vetter et al. (2024) observed, our capacities for critical and ethical use of AI can only emerge “through active negotiation, practice, and dialogue” based on “contextual understanding” of embedded roles in discourse communities and response to unique rhetorical challenges (p. 9).

Writing with AI as identity development and learning

Learning to write involves not only skill acquisition but also identity formation (Scott et al., 2015). The threshold concept “Writing Enacts and Creates Identities and Ideologies” (TC 3), addresses this developmental link between identity formation and writing skills. Writers develop skills through sustained engagement with discourse communities, gradually internalizing the conventions, values, and ways of thinking that characterize disciplinary practice. This process proves transformative: “As people progress through their major discipline(s), writing increasingly complex texts in the process, they are also writing themselves into the discipline(s)” (p. 56). It is also a reciprocal process in which writers draw “on personal knowledge and lived experience in creating new texts” (p. 54), while the act of writing simultaneously renegotiates and extends that knowledge. Writing becomes the primary means through which students construct professional and disciplinary identities, bringing their own histories, knowledge, and values into productive dialogue with the knowledge-making practices that define their chosen fields.

This developmental arc connects to a closely related threshold concept: “All Writers Have More to Learn” (TC 4). As Rose explains, “there is no such thing as ‘writing in general’; therefore, there is no one lesson about writing that can make writing good in all contexts” (Rose et al., 2015, p. 60). Writers must continuously adapt established practices to novel situations. Such adaptation demands “different kinds of practice, time, and effort,” a threshold concept that proves troublesome because it “locates writing specifically as a practice situated within communities, which suggests how complex writing is and how, as an activity, it spans a lifetime” (Rose et al., 2015, p. 65). This adaptive capacity develops through externalizing texts for critical examination, embracing failure as essential to growth, engaging in iterative revision, and cultivating the metacognitive awareness necessary for self-assessment.

The threshold concepts of identity formation and continuous learning complicate binary framings of AI as either a benign productivity tool or a threat to resist. Because writing development depends on sustained cognitive effort, rhetorical judgment, and the gradual formation of disciplinary identity through practice, AI can genuinely support or undermine those developmental processes depending on how writers engage it. Wang and Wang (2025) argued that transactional approaches—positioning AI as something writers should simply use or avoid—fail to account for how meaning emerges through what Gries (2019) described as “an assemblage of bodies, technologies, materials, and so forth [that] intra-act with students to produce various assemblages of texts, images, and other artifacts” (as cited in Wang & Wang, 2025, p. 3). From this perspective, AI represents an intensification of the relational dynamics that already characterize technologically mediated composing; cultivating AI literacy is “essentially, relational work rather than transactional work” (Wang & Wang, 2025, p. 4). Embracing this relational complexity is essential for navigating “distributed agency” productively (Wang & Wang, 2025, p. 4).

Writing instructors perceive this tension in pedagogical practice. In a survey-based study of first-year composition instructors, Pandey et al. (2025) found that instructors worry AI tools “undermine students’ creativity by prompting them to rely on AI-generated content leading to erosion of rigor needed for academic strength and originality of writing” (p. 9). Yet instructors also recognized AI’s potential for supporting rather than replacing student learning: assisting with brainstorming, helping overcome writer’s block, providing feedback on structure and clarity. This ambivalence reflects the relational complexity Wang and Wang (2025) identified: when students automate cognitive work or surrender strategic decision-making to AI, they risk failing to develop the capacities that sustained practice builds; when they engage AI as collaborative resource while maintaining rhetorical ownership, they may extend their composing capabilities and learning development. The pedagogical challenge is helping students navigate this distinction.

Multimodal composition scholarship offers productive frameworks for addressing this challenge. As Jiang (2024) argued, when students make deliberate design choices across modes and media, they develop heightened metacognitive awareness of how composing processes serve rhetorical purposes. Multimodal work engages writers in a recursive process of “navigating available designs, designing, and redesigning” (Jiang, 2024, p. 5). Jiang (2024) argued that AI-assisted multimodal composition can similarly promote strategic decision-making and the exercise of rhetorical agency that builds metacognitive ownership of composing processes. However, multimodal contexts do not automatically resolve the challenges of working with AI. Students still face “wicked problems” where requirements conflict, solutions remain partial, and processes constantly evolve (p. 4). Success requires strategic choices that keep outputs aligned with rhetorical purposes; these are choices that demand human judgment about disciplinary expectations, audience needs, and communicative goals.

These three dimensions of disruption—cognitive, social/rhetorical, and developmental—identify challenges that students must navigate when learning to write with AI. Writers must manage cognitive load while maintaining metacognitive control, assume responsibility for rhetorical and disciplinary knowledge that AI cannot access or evaluate, and develop skills and identities as writers by making strategic choices while navigating distributed agency. What conceptual transformations enable students to navigate these interconnected challenges productively? The methodology that follows describes the learning design and analysis procedures through which we investigated that question.

Methodology

Study context and participants

This study examines student reflections from an elective “AI and Writing” course taught at a public midwestern university across two iterations in Fall 2023 and Fall 2024. The course attracted 38 undergraduate students from diverse academic backgrounds, spanning 22 different majors across engineering, arts and humanities, business, and sciences. This population is well suited to exploring how threshold concepts in AI-assisted writing emerge in mixed-expertise learning communities.

In our previous study, we offered an in-depth examination of the learning design and how it integrated a holistic approach to

promoting AI literacies—including functional, critical and ethical, and creative dimensions—with a scaffolded approach for self-regulated learning skills (Anders & Dux Speltz, 2025). In this second study, we address conceptual transformations, examining how students articulated their changed self-understanding of writing with AI practices. Our analysis focuses on students' final course reflection essays; this distinct data set offers unique perspectives on how students navigated the troublesome liminal thresholds articulated in our literature review.

Learning experience design

The course employed a two-phase structure integrating scaffolded practice with self-directed application. During the first ten weeks, students completed weekly “creative challenges,” which were structured experiential activities designed to surface conceptual obstacles while building AI literacy competencies. Taking inspiration from Tate et al. (2023) framework, the course was organized around five learning objectives: (1) understand how generative AI and large language model technologies work, including their capabilities and limitations; (2) analyze ethical challenges of using AI for writing and learning; (3) find and use AI tools for specific purposes, genres, and media; (4) create effective prompts and use strategies to find accurate information; and (5) compose texts that integrate AI-generated content (Anders & Dux Speltz, 2025). The creative challenges were sequenced to address these objectives while deliberately targeting problematic assumptions that students might bring to AI interaction (see Table 2 for examples).

The first set of creative challenges addressed the tendency to conflate linguistic fluency with cognitive competence, which Mahowald et al. (2024) identified as a persistent feature of how people interpret AI outputs. The “Create a Fluent Hallucination” activity, inspired by Mollick (2023), asked students to deliberately generate plausible but inaccurate outputs, including fabricated events and invented sources, to examine how AI can produce superficially convincing but incorrect information. By positioning students as producers rather than consumers of misinformation, the activity made the risks of uncritical AI reliance concrete and difficult to ignore, establishing the need for sustained human attention, fact-checking, and evaluation as foundational practices.

A second cluster of activities was designed to develop evaluative judgment grounded in rhetorical situations and disciplinary knowledge. A “Prompt Competition” challenged students to craft persuasive messages for specific audiences using competing approaches: simple zero-shot prompts versus structured processes integrating audience analysis, rhetorical strategy, and iterative refinement. By requiring students to conduct rhetorical analysis, select persuasive strategies, gather supporting research, and use these elements to guide AI-assisted drafting, the activity made visible how disciplinary knowledge can substantially improve the quality of AI-assisted writing. A follow-up sequence introduced advanced prompting strategies that students applied and compared across varied contexts (Dux Speltz & Anders, 2024). Additionally, an “AI Ethics Tutor” engaged students in scenario-based interactions designed to develop their capacity for evaluating and responding to situated ethical dilemmas in AI use moving them from reliance on fixed rules toward contextual and principles-based judgment (Anders & Dux Speltz, 2023).

A third set of challenges was designed to address distributed agency in AI-assisted composing. The “Create an AI Assistant” challenge required students to design a workflow for a task they found difficult, explicitly identifying which tasks the human would handle and which the AI would support (Anders & Dux Speltz, 2024). Students first analyzed their chosen problem, identified sub-problems and existing tools, and then created a detailed plan specifying human and AI roles. This process surfaced gaps in students' own process mapping and overestimations of AI's capabilities, requiring them to iterate their way toward a workable division of labor. Together, these activities created iterative opportunities for students to encounter the situated constraints of AI interaction and develop more intentional, human-directed approaches to collaboration.

The final five weeks shifted from scaffolded challenges to self-directed projects requiring students to identify an academic, professional, or personal challenge and develop a multimodal deliverable using at least two distinct AI applications. Projects were supported through a structured sequence of planning, prototyping, peer feedback, and a final presentation designed to function as a cumulative mastery experience. This offered an opportunity for students to apply their developing AI literacies in an authentic and personally meaningful context. Students' final deliverables spanned diverse categories: interactive applications integrating

Table 2
Examples of creative challenges.

Learning Objective	Creative Challenge	Purpose
Understand AI capabilities and limitations	Create a Cut-up Text	Explain how large language models generate text by working through an analog next word prediction activity that illustrates the underlying process
	Create a Fluent Hallucination	Examine how AI produces convincing but incorrect outputs, identifying the specific conditions that can lead to outputs with factual errors
Analyze ethical challenges of AI-assisted writing	Use an AI Ethics Tutor	Evaluate different approaches to situated ethical dilemmas for AI use, developing and justifying personal principles for responsible practice
Find and use AI tools for specific purposes	Choose a Specialized Writing Tool	Compare tools designed for specific writing tasks and assess which best fits a given purpose, audience, and context
Create effective prompts and find accurate information	Prompt Competition	Investigate how prompt design and information shapes AI outputs by testing and comparing different approaches, drawing conclusions about what produces better results and why
Compose texts that integrate AI-generated content	Create your own AI Assistant	Design a workflow for a situated task that explicitly assigns roles to human and AI, constructing a process that keeps human judgment central to planning and outcomes
	Use Creativity Techniques to Solve Problems	Apply creative problem-solving frameworks with AI support to generate and develop ideas for a personally meaningful challenge, directing the AI toward contributions that serve specific goals

disciplinary expertise with AI capabilities, such as a sustainable seed selection tool for farmers; creative multimedia works combining AI-generated text and imagery; media productions using AI for rapid prototyping and rhetorical analysis; and writing projects addressing specific purposes, genres, and audience needs.

Data and analysis

Our analysis examines students' final course reflection essays, submitted during the last week of each semester. These reflections were guided by prompts asking students to articulate their evolving understanding of AI-assisted composition in four areas: their most valuable technical learning and plans for future application; specific examples of integrating AI with disciplinary or personal expertise; their developing sense of balance between human and AI capabilities; and their principles for ethical AI use going forward.

We used Meyer and Land's (2005) five threshold concept characteristics as sensitizing criteria: troublesome, bounded, transformative, integrative, and irreversible. This meant attending to moments where students contrasted earlier assumptions with current understanding, expressed surprise at what productive AI collaboration actually requires, or described shifts in perspective they could not imagine reversing. We also looked for how transformed understanding was enacted in situated applications, including instances where students demonstrated conceptual frameworks in action through specific practices in their own writing and composing work. The analysis process involved independent review and coding, followed by shared discussion to reconcile interpretations and develop consensus on emergent patterns. Through this collaborative analysis, we identified key threshold concepts for writing with AI that recurred across students' reflections, as expressed through each student's unique personal experiences and disciplinary perspectives.

Findings: Threshold concepts for AI and writing

Analysis of students' final reflections revealed three interrelated threshold concepts characterizing productive human–AI collaboration for writing with AI (see Table 3). The subsections that follow examine each in turn, tracing how students moved from pre-liminal assumptions through liminal struggle toward postliminal reorientation, as they encountered tensions, revised their understanding, and articulated insights that signaled threshold crossings.

Writing with AI is an experimental process

The first threshold concept describes a shift in what productive engagement with AI requires, extending the foundational understanding of writing as a cognitive activity (TC 5). A preliminary orientation visible across reflections is a tacit expectation for ease of use and for AI to work as an enhanced search tool, capable of returning usable results from simple queries. From that starting point, the reflections trace an incremental working-out of why the orientation falls short: a recognition of the limits of single-prompt exchanges, then experimentation with more detailed prompts integrating contextual information, and finally a gradual recognition of iteration itself as a central logic of productive use. In the postliminal phase, experimentation takes shape as a deliberate strategy using a plan-and-test cycle in which prompts, outputs, and revisions are directed and evaluated against the writer's own goals. Table 4 maps this movement across preliminary, liminal, and postliminal phases, and the snapshot that follows traces one learner's experience of this threshold in practice.

Snapshot 1: Writing with AI is an experimental process

To examine how a student came to reconceptualize AI engagement as an experimental process, we turn to the experience of an English major from the Fall 2024 course who used generative AI to support an extended fiction project. This student's reflections trace a movement from initial, ineffective engagement with AI toward a more deliberate, iterative practice grounded in planning and self-directed evaluation. Their account also illustrates how this conceptual shift does not eliminate the tension between cognitive off-loading and metacognitive direction setting; instead, it makes that tension a point of ongoing awareness and focus.

Before the course, the student's approach to AI reflected an implicit assumption that the tool would handle the work if given sufficient raw material: "I used to throw a jumbled bit of information at Chat GPT and it wouldn't give me the output that I wanted." Throwing information at the tool describes an unstructured transfer rather than a directed process; the student was not yet designing or sequencing the interaction but testing whether the tool could complete the task on its own. The liminal thresholds section identified

Table 3
Threshold concepts for writing with AI.

Concept	Insight	Representative Quote
Writing with AI is an Experimental Process	Writers engage in experimental iteration and deliberate practice rather than one-time transactional exchanges	"I realized that the true power of AI lies in the ability to work iteratively and collaboratively with the system in an ongoing back and forth dialogue."
Writing with AI Requires Expertise and Dialogue	Writers exercise their own situated rhetorical understanding, disciplinary expertise, and evaluative judgement rather than deferring to authoritative sounding outputs	"Because of my background as a data science major, I knew exactly what to ask the AI to generate and I knew how to verify the answer correctness."
Writing with AI Should Augment Rhetorical Agency	Writers maintain strategic oversight and make choices to augment their capabilities and effectiveness rather than replacing their intellectual and developmental effort	"My ideas and creativity are the base for everything, AI is just the tool I use to execute my ideas."

Table 4

Thematic insights for “Writing with AI is an experimental process”.

Thematic Insight	Representative Quotes
<i>Preliminal:</i> Discovering limitations of treating AI as a search tool for quick queries, with a tacit expectation that it should require little effort.	“Before this class, I had only been using GPT or other LLMs as a ‘Google replacement’ instead of using it for what it really is: a chatbot meant to be conversed with.” “Before this class I had never worked with AI aside from asking a quick question.”
<i>Liminal:</i> Beginning to see that output depends on input, which shifts effort upstream into planning and providing context.	“From personal experience and what we did and covered this semester in class, it’s clear that the output of almost all AI tools/systems is dependent on the quality and/or quantity of the human input. Most of the time, the desired output may take several attempts to achieve, if at all.” “I’ve found that the more information I give it, the more it’s likely to be correct ... Putting in enough effort to help AI help you, is so worth it, because if your goal is to get assistance, why not give the assistant the tools it needs.”
<i>Postliminal:</i> Recognizing experimental iteration as a central logic of productive AI use and enacting it in plan-and-test approaches.	“One of the most valuable techniques I have learned in this course is using trial and error with AI prompting. This approach has allowed me to refine my prompts iteratively, leading to more accurate and relevant outputs from AI tools.” “Iterative prompting involves refining and adjusting prompts based on initial outputs to achieve higher-quality results. ... For example, when drafting an outline for an argumentative essay, I initially prompted the AI to generate a basic structure. I then revised my prompts to request additional supporting details, counterarguments, and examples specific to my thesis. This iterative process improved the coherence and depth of the essay.” “Prior to this course, I simply would have prompted it and taken its first answer. Now, I will be focusing on getting a better answer... by refining it with an iterative approach.”

this as an intuitive, though troublesome, starting assumption that AI should function as a more capable information retrieval tool, available to handle rhetorical labor on demand.

Reflecting on their learning experiences, the student identified prompting as the skill that most changed their relationship to AI: “Of everything that I have learned in this class, prompting skills has definitely been the most valuable skill. It has given me the ability to efficiently use AI.” Prompting well requires the writer to clarify what they want before they ask, shifting cognitive effort upstream from reacting to outputs toward planning inputs. This is an early movement toward the active experimentation that this threshold concept describes, though expressed here in fairly general terms.

The student extended this logic further through a technique they learned in class called “flipped interaction prompting,” in which they reversed the typical direction of the exchange: “My favorite technique is flipped interaction prompting, where the AI asks me the questions... By explaining my own plans, it helps me grasp exactly what I’m trying to do.” The student redirects AI from content generation toward supporting reflection on their own goals and intentions before committing to a direction. The exchange becomes a structured way to examine one’s own thinking before acting on it; it is an approach to planning that treats clarifying intentions as the necessary prior work of productive AI engagement.

The student did not leave this as an isolated technique. They connected it to the social practice of testing ideas through peer dialogue: “As a writer, I like to share my work with my friends... I call it ‘bouncing an idea off them’... I find that having conversations with AI is a good replacement.” They immediately clarified that “it’s not the same as human interaction by any means.” This move from a specific technique toward a broader principle reflects the kind of metacognitive abstraction Adler-Kassner et al. (2012) describe as characteristic of developing threshold awareness: a learner begins to recognize the work that a particular practice is doing, not just how to do it. The student places flipped prompting within a larger family of strategies for using external dialogue to clarify thinking, the same function that peer feedback and community of practice discussion served in the course more broadly. The explicit caveat about AI’s limits keeps the analogy grounded, reflecting a critical orientation toward what the technology can and cannot offer.

The student’s most fully developed enactment of these strategies came in their final project, where they applied them to producing an extended piece of fiction. At this stage, the student describes a reoriented perspective on difficulty itself: “It took a lot of iterative improvement to make sure that the writing did not sound bland. I made sure to be clear and detailed in my prompts... Once it was written, I made sure to tell the AI what it had gotten right... Then I would tell it what it did wrong, and I would suggest how to fix it.” The student is describing a sustained cycle of evaluation and redirection, guided by a stable sense of what the writing needed to achieve. Where this type of friction had initially signaled a failure of tool and task, it has become the focus of the work itself, managed against the student’s own established criteria. This demonstrates a mindset of active experimentation being enacted: the student measures progress against their own standards, rather than adjusting standards to accept whatever the AI produces.

Even at this stage, the student remained alert to the ongoing risk of ceding creative control: “I don’t like when AI takes over the creative process for me, because I find that my creative mind is more dynamic and ideas more unique than what AI can come up with.” Across the full arc, this account illustrates a gradual process of a learner recognizing that experimentation and relying on their own ideas is the foundation of composing effectively with AI. From initial ideation, to developing structured prompts grounded in clear plans, to iterating and evaluating outputs against their own goals, this student came to define for themselves what effective experimentation requires in practice. Undoubtedly, writers will continue to experiment with how much cognitive effort to offload, both to free up higher-order thinking and to reduce effort in general. Individual students may be tempted to cut corners on how much they offload and how carefully they direct the process. The point, however, is that once acquainted with iteration as a method, the trade-offs become visible. The question is no longer just how much to let AI handle but how to direct the process through deliberate and

evaluative choices. A learner who arrives at this understanding has crossed a threshold: where they once issued a single prompt and accepted what came back, they now begin by asking what they want to achieve and how to plan and experiment toward it.

Writing with AI requires expertise and dialogue

The second threshold concept reframes the role of situated expertise in AI-assisted writing, extending disciplinary understandings of writing as a social and rhetorical activity (TC 1) that relies on situated genres, conventions, and forms (TC 2). Across students’ reflections, this liminal journey begins with identifying troubling gaps that disrupt initial illusions of AI competence including outputs that sound fluent and authoritative but miss factual details or fail to fit the contextual demands of specific writing tasks. This recognition surfaces the writer’s own tacit situated rhetorical and disciplinary knowledge as the resource that must be developed and communicated to guide both planning and evaluation. In the postliminal phase, supplying that knowledge becomes a practice in its own right: the collaborative process serves as a means for discovering, articulating, and refining the writer’s own thinking. Table 5 maps this movement across preliminal, liminal, and postliminal phases, and the snapshot that follows traces one learner’s experience of this threshold in practice.

Snapshot 2: Writing with AI requires expertise and dialogue

To examine how a learner encounters the role of expertise and dialogue in AI-assisted writing, we turn to the experience of a journalism student from the Fall 2024 iteration of the course. This student’s account illustrates a movement from assuming that AI can handle genre-specific writing tasks toward recognizing that their own disciplinary knowledge is what makes productive collaboration possible. The movement is neither immediate nor frictionless; it emerges through a specific failure and unfolds across successive adjustments.

The student’s initial approach to AI reflected what Mahowald et al. (2024) identified as a persistent feature of how people interpret AI outputs: the tendency to assume that formal linguistic fluency signals functional competence. Early experimentation surfaced this assumption directly: “I’ve messed around a bit with ChatGPT helping with writing leads, and ChatGPT doesn’t really follow the desired structure for a lead or standard journalism rules.” The student begins with an informal testing orientation, trying to see whether AI could handle the task rather than starting from a clear sense of what the task required. The disruption was not simply that AI produced errors but that fluent, contextually plausible output failed against disciplinary standards. Formal linguistic competence, in other words, does not extend to the situated understanding of what effective journalism requires for this genre and professional context.

This failure does something important before any corrective action occurs: it makes the student’s own disciplinary knowledge visible as a resource. Recognizing that the output “doesn’t really follow the desired structure” presupposes that the student has a structure in mind and through their learning they have developed criteria they can apply, even if they have not yet identified them. The failure does not only expose AI’s limitations; it surfaces knowledge the student possessed but had not yet directly enacted. At this moment, the student begins to see the gap between fluency and competence and, in turn, to reckon with what they themselves know.

A later interaction shows how that recognition can become productive. Describing a subsequent use of AI, the student explains: “When I used it to help give me feedback, most recently I told it all of the knowledge I had about writing a lead and the structure that I was choosing to follow. I crafted a brief knowledge base for it to follow and then shared my already written lead with it and it was able to give me much more accurate feedback.” The first move is a significant departure from “messaging around.” Before engaging AI, the student explicitly assembles their disciplinary knowledge, identifying the conventions that govern a lead, and creating a plan for the rhetorical strategy they intend to follow. In Bearman and Ajjawi’s (2023) terms, this reflects discipline-specific evaluative judgment treated as something the writer must supply; and in this account, planning and evaluation are two sides of the same expertise. The disciplinary knowledge that shapes the plan also supplies the criteria for judging what AI returns. This also reorients how the student

Table 5
Thematic insights for “Writing with AI requires expertise and dialogue”.

Thematic Insight	Representative Quotes
<i>Preliminal:</i> Encountering the gap between AI’s fluency and both its factual accuracy and contextual fit for specific writing purposes.	“Before taking this class I would have had no idea whether or not AI was doing a poor job, or a good job. I could have asked it to quote an article or write an essay using a certain source and it could have made stuff up completely. Now, I know what red flags to look for when using AI.”
<i>Liminal:</i> Using one’s own situated rhetorical and disciplinary knowledge to guide planning and evaluation of AI-assisted work.	“I often started my conversations with AI with enough knowledge base to get both of us on the same page. I preferred that to ‘random thought’ AI, as I liked to call it.” “Being able to question myself about my world, characters, and arcs is critical... I can see myself using generative AI tools to begin the worldbuilding process or integrate into the development process.”
<i>Postliminal:</i> Using the collaborative process to discover, articulate, and refine one’s own knowledge and thinking, a practice essential to productive AI use and generative for the writer’s understanding.	“Personally, I don’t believe AI could make a compelling creative story on its own. If you ask it to write a fictional story, it comes out with a very bland excerpt that is usually unoriginal. But you can create a compelling and unique story if you give it expert or personal knowledge.” “When prompting the AI to simulate evaluating an aircraft’s wing design, I included parameters like aerodynamic efficiency, material weight, and cost constraints. This level of detail allowed the chatbot to generate realistic and actionable outputs.”

engages AI: rather than asking it to generate a lead, they draft one themselves and ask for feedback against the goals and criteria they have just articulated. In Vetter et al.'s (2024) terms, it reflects an approach to collaboration in which the student assumes rhetorical leadership, in this case, by first producing the initial work that AI then helps develop.

The student does not leave this as a technique for one task. Reflecting on an earlier use of AI for brainstorming, they describe a closely related approach: "Referring back to my news story I am writing for another class, I used AI to help me with my brainstorming process by giving it the guidelines of my assignment and explaining the ideas I already had... This allowed me to share my ideas and the knowledge I already had about my topic and then AI helped me refine my idea in order to get a strong start on my project." The task involves generating starting ideas rather than evaluating a drafted text, but the underlying move is the same. In both instances, the student begins by assembling what they know and making it available to AI. In the lead-feedback example, they provided the conventions of a journalistic genre. In the brainstorming example, they provided assignment guidelines and ideas they had already developed. The student is beginning to describe a broader principle for bringing their own expertise actively into the exchange. Their own disciplinary and subject knowledge, which previously may have functioned tacitly as background preparation, becomes the active resource that directs AI and supplies the criteria by which its output can be judged. Articulating tacit knowledge helps the writer develop and clarify their own thinking. Hence, making one's expertise explicit not only compensates for AI's limitations but also serves the writer's own development.

Read as a whole, the account also shows that this principle emerged dialogically. The student's recognition that their own knowledge and ideas must be made explicit before AI can contribute usefully did not precede their use of AI; it grew out of "messing around" first and noticing where the output fell short. Different writers will likely to arrive at different approaches: how much disciplinary context to supply before engaging AI, at which points in a task the writer's evaluative criteria most need to be articulated, and where their own thinking most needs protection from premature refinement. In this account, a learner begins to see the need to articulate their own knowledge not as a limitation imposed by AI collaboration but as one of its productive features. The collaborative process becomes an occasion for developing, clarifying, and refining the thinking that the writer then uses to direct AI. A learner who arrives at this recognition has begun to cross a meaningful threshold; it is one that reframes articulating expertise as both essential to productive AI use and generative for the writer's own understanding.

Writing with AI should augment rhetorical agency

The third threshold concept addresses how writers navigate conditions of distributed agency, extending foundational understandings of writing as involving identity formation (TC 3) and as a site of continuous learning (TC 4). The temptation to use AI to fully automate the work and thinking of writing is a concern voiced by educators and echoed in students' own reflections. In the preliminal phase, learners encounter the quality and process risks of letting AI carry too much of the work: outputs that fall short of the writer's standards, and composing processes in which the decisions that most shape the result are made by the tool rather than the writer. The liminal phase involves experimenting with how to assign AI different roles and how to structure the process, identifying where to plan ahead, where to supply ideas, and where to step in so that the choices that matter remain the writer's own. In the postliminal phase, writers come to identify where their own ideas and judgment matter most in composing processes and shape their use of AI around those moments. Crossing this threshold means asking whether the process is adding to or subtracting from one's own thinking and contributions, and how that difference shapes the quality of both writing outputs and rhetorical outcomes. Table 6 maps this movement across preliminal, liminal, and postliminal phases, and the snapshot that follows traces one learner's experience of this threshold in practice.

Snapshot 3: Writing with AI should augment rhetorical agency

To explore how a learner comes to understand writing with AI as a matter of augmenting rather than automating rhetorical agency,

Table 6
Thematic insights for "Writing with AI should augment rhetorical agency".

Thematic Insight	Representative Quotes
<i>Preliminal:</i> Encountering the limits of transactional framings that cast AI use as either full automation or full avoidance.	"Before taking this class, AI always felt taboo in a way. Using AI always felt like a way around doing the work, cheating." "I value my own work too much to use AI consistently. Just the use of AI at all is a process I'll now have that I'll take from this class."
<i>Liminal:</i> Experimenting with approaches to human and AI roles and interaction, and finding that those choices surface the writer's skills, values, and purposes.	"I think it gave me a better sense of my skills as a writer, knowing what I could write and what I couldn't." "One big lesson I've learned is that being involved every step of the way makes the final result better. The AI can handle repetitive or time-consuming tasks, but it's my creativity and decision-making that make the output meaningful."
<i>Postliminal:</i> Making strategic rhetorical and process choices that use AI to extend the writer's own composing capacities and rhetorical agency.	"The balance that I find most effective is by using AI as a tool in the beginning and end of my process but leaving the middle up to me." "I had to use multiple AI tools to learn how to make a chatbot effectively ... It was very easy to put every question I had into ChatGPT to get a quick answer even if the question wasn't complicated and it started to be a crutch for me. I had to set restrictions for myself on what I should and shouldn't ask AI for help with and it made finishing my project a lot easier and feel more like something I made."

we turn to the experience of a business analytics student from the Fall 2023 iteration of the course who reflected on an essay analysis project for their final deliverable. This student's account traces an incremental working-out of how to distribute labor between themselves and AI, moving from an initial recognition that the tool's default behavior threatened both process and outcome, through a series of deliberate redesigns that placed their own ideas and choices at the decision points that most shaped the work. The account shows how distributed agency is not a single orientation to be adopted but a practice assembled incrementally across a project.

The student's initial, preliminary engagement surfaced a dual dissatisfaction with AI's default mode of responding to an opening prompt:

"When it came to writing the essay I initially wanted to submit as part of my final project, I knew a decent amount about the subjects of each one. The hardest part of this, regarding the AI, was not letting it fill in aspects of the character for the story-telling process that I knew well. After my initial prompt about the essay, Chat GPT started to develop its own version of what I detailed I wanted. Not only did I want to write the majority of the essay myself, but its descriptions were not very detailed or well explained."

Two kinds of knowledge meet in this moment. The student has tacit subject knowledge, "I knew a decent amount about the subjects," and emerging knowledge about how AI tends to behave when given a general prompt. The complaint that follows concerns both output quality and what AI's default mode does to the composing process. The tool's tendency to "develop its own version" displaced exactly the authorial work the student wanted to retain, while also failing to produce descriptions that met their standards. In Knowles's (2024) terms, the student has begun to sense the risk of letting AI carry too much of the rhetorical load, as a preliminary step to thinking about how the load might instead be shared. This preliminary moment also registers a growing awareness that process and outcome are connected and that addressing them both requires a coordinated strategy.

Shifting into liminal enaction, the student began using their own knowledge to plan, developing an approach that paired strategic choices about content with specific guidance for how AI should follow and implement the plan:

"After seeing how it processed my initial prompt, I altered it to include that I wanted it to help me develop an outline versus a draft. The outline needed to be broken down into several specific categories, sub-categories, and include marked spaces for related visuals. I also included a clause to say I wanted it to add the notes we would collect into the correlation subcategory and if it applied to multiple, ask and put it only into one."

The student also changed their approach to rhetorical load sharing by asking AI to generate an outline rather than a full draft. This reassigns AI's contribution from expression to supporting arrangement and breaks the work down into smaller iterative stages that allow for more granular prompting and feedback. The outline's detailed specifications for categories, subcategories, and visuals are all decisions the writer made in advance, drawing on their own sense of what the project required. Restructuring the process in this way is a deliberate move away from full automation and toward an interaction pattern that keeps the writer's thinking and direction at the center of the work.

The student extended this logic from planning into coordinating the collaboration itself, redesigning the interaction so that their ideas and choices would be threaded through each step: "In order to allow me as the user to supplement its knowledge with my own, I asked it to complete every step as an interview process. Using this method, the individual character analysis had better depth regarding the characters' interpersonal relationships throughout the show's entirety." In this move the logic of rhetorical load sharing is used to define an interaction pattern with the AI that emphasizes the learner's thinking process. An interview format turns a single prompt-and-output exchange into a sequence of moments at which the writer supplies knowledge and makes choices. The phrase "supplement its knowledge with my own" identifies this shift in orientation: the writer's contributions are not delivered once up front and then handed off but distributed across the encounter. The outcome claim, that the analysis had "better depth," matters less as evidence that the redesign succeeded than as evidence that the student is evaluating their process and its outputs against their own substantive standard for the work. Read this way, depth of analysis becomes a criterion that can help the writer evaluate whether the collaboration is legitimately serving to augment their capabilities.

The student also extends this approach beyond the single project. Reflecting more broadly on their composing practice, they articulate it as a general principle for coordinating AI collaboration: "Another method I found useful was the process of outlining a mega prompt ... in which you detail the problem/project you're working on, the kind of outcome you're looking to get at the end, and the role of the machine in the process." The three components — identifying the problem, describing the intended outcome, and specifying AI's role — correspond directly to the moves traced across the preceding account: drawing on subject knowledge before engaging AI, holding evaluative criteria in view, and deciding in advance how the rhetorical load will be distributed. The student has articulated a reusable strategy for directing human-AI collaborations that can be carried forward into new projects.

This reflective account shows how distributed agency can be enacted through strategic rhetorical and process choices: decisions about how the work is structured, how the interaction unfolds, and what the writer supplies at each stage so that their own knowledge and judgment lead the composing. The particular approaches this student arrived at are one instance of a broader set of choices available to writers, which may vary considerably based on the rhetorical situation and the writer's own values and needs. As we have noted, writers will experiment with how much of the composing work to delegate, and individual learners may be tempted to skip past the decisions that most shape the work. However, crossing this threshold means recognizing that a human-centered composing process, one that integrates the writer's own knowledge, insights, and judgment, shapes both the writer's development and the quality and impact of the work.

Discussion: Toward an ethics of experimentation

As Vetter et al. (2024) and Wang and Wang (2025) have argued, responsible AI use cannot be reduced to fixed prescriptions. Cultivating AI literacy is relational work requiring ongoing dialogue within specific rhetorical contexts. An experimental approach is more than a practical necessity for engaging technologies whose affordances, constraints, and impacts remain incompletely understood; it represents an ethical stance toward our pedagogical responsibilities and disciplinary research. This study demonstrates what such engagement makes possible. By designing a community of practice where students navigated AI-assisted composition through scaffolded challenges and self-directed projects, we observed and documented emergent challenges, practices, and conceptual transformations.

Our three emergent concepts extend and transform the foundational threshold concepts established in *Naming What We Know* (Adler-Kassner & Wardle, 2015). Part of what gives the original threshold concepts their continued relevance is that they name perennial tensions that writers must navigate throughout their development and continually renegotiate as they encounter new contexts, genres, and technologies. Writing with AI represents a new frontier in this process of development and transformation.

The threshold concept “Writing with AI is an Experimental Process” names a fundamental reorientation: productive engagement requires iterative refinement rather than transactional exchange. This insight connects to a perennial challenge framed as “Writing Is (Also Always) a Cognitive Activity” (TC 5). Learning new skills introduces cognitive load and can degrade short-term performance, even as routinization eventually frees attention for higher-order concerns (Dryer et al., 2015). The same dynamic holds for experimental engagement with AI with one important distinction. In prior accounts of writing as cognitive activity, routinization was largely a passive process: practiced skills receded from conscious attention through accumulated experience. With AI, offloading cognitive effort becomes a continuous deliberate choice, which means that experimentation requires metacognitive awareness of the trial-and-error process itself and active direction toward one’s own goals rather than accepting whatever the process produces. Writers will necessarily struggle through failed prompts and ineffective interaction patterns before identifying approaches that work for specific tasks. Yet those who understand experimentation as a method can persist through initial friction, developing repertoires of prompting strategies and workflows suited to their purposes. Over time, effective patterns become routine, creating cognitive bandwidth for the higher-order rhetorical and strategic thinking that enables effective writing and composition practices.

The threshold concept “Writing with AI Requires Expertise and Dialogue” captures a second reorientation: writers must develop and provide relevant expertise rather than deferring to AI’s apparent authority. This insight connects to the perennial challenges that writing is a social and rhetorical activity (TC 1) and that writing speaks to situations through recognizable forms (TC 2). Effective communication requires situated knowledge of audiences, purposes, and disciplinary contexts with genres and conventions providing the primary means through which writers recognize and respond to the demands of specific rhetorical situations. Because AI can generate text that mimics these features without possessing the underlying understanding they normally signal, writers cannot rely on fluency as evidence of quality. This makes rhetorical and genre analysis skills even more imperative: writers need to be able to articulate audience knowledge, disciplinary context, and genre criteria before engaging AI, because it is the only way that AI can be productively directed and its outputs meaningfully evaluated. Through dialogue, writers discover both where AI falls short and what expertise they themselves need to develop and communicate. The process of assembling adequate context, relevant information, and appropriate guidelines can deepen a writer’s grasp of the social and rhetorical dimensions of their task. In this sense, writing with AI can become a generative process for developing and applying situated social and rhetorical insights for communicative practice.

The threshold concept “Writing with AI Should Augment Rhetorical Agency” names a third reorientation: in conditions of distributed agency, writers must make strategic choices about what work to delegate and what to retain. This insight connects to the perennial challenges that writing enacts and creates identities (TC 3) and that all writers have more to learn (TC 4). Writers develop skills and identities through sustained engagement with discourse communities; they “write themselves into disciplines” through the difficulty of mastering disciplinary discourse (Scott et al., 2015). Because there are no general-purpose writing skills, each new context requires patient rediscovery of which approaches transfer and which must be adapted (Rose et al., 2015). Generative AI intensifies this challenge in a specific way: because AI can handle execution across virtually any stage of the composing process, writers face a genuine risk of automation in which they let AI make the choices and do the work that most shape both the quality of what they produce and the development of their own capacities over time. Moving through this threshold means recognizing that within any composing process there are key moments where articulating one’s own knowledge and making discerning rhetorical choices are most consequential, and that identifying and protecting those moments is what keeps the collaboration augmentative rather than substitutive. As strategic choices accrue and writers learn to map the points where their judgment most matters, they can gain new perspective on their own capabilities, develop more intentional approaches to rhetorical load sharing, and produce work that more fully serves their own communicative purposes.

These threshold concepts will not eliminate the tensions between AI uses that augment writing development and those that undermine it, nor should they be expected to. Together, they offer a more precise account of where those tensions are located: in the metacognitive demands of experimental process, in the rhetorical and disciplinary knowledge that productive dialogue requires, and in the strategic choices that keep composing in service of the writer’s own purposes and development. Students who cross these thresholds do not fully resolve the tensions between augmentation and automation, but they gain greater awareness and conceptual resources to navigate them deliberately and strategically.

Limitations and future directions

This study documents threshold crossings as articulated in students’ end-of-course reflections within a specific experimental

learning context. The findings open several productive directions for future research. Our goal was to identify patterns in the troublesome and transformative experiences of students learning to write with AI. Self-reported reflections are appropriate for this purpose because they show how students articulate and begin to apply the lessons of their own experiences. At the same time, this scope invites future research using complementary methods. To examine the durability of changed perceptions, longitudinal studies could trace how student practices evolve over time and whether the threshold concepts we identify prove durable over time. Given the stakes for how AI can augment or erode writing capabilities, future work could also examine how writing with AI affects the quality of student writing and multimodal composing products, comparing how student progression along the liminal spectrum of these concepts corresponds to gains in writing skills and performance. Future studies could also examine the full range of student trajectories, including partial or resistant engagement, and how different course designs and pedagogical practices help students cross these thresholds.

In addition to identifying an initial set of threshold concepts for AI-assisted writing, this study offers a methodological contribution. We demonstrate how communities of practice, observed through experiential and reflective practice, can help our field understand shifting learning challenges during periods of technological transformation. We approached the course experimentally out of necessity, as generative AI emerged too rapidly for established pedagogical frameworks to guide our practice, and from conviction that an ethics of experimentation offers a compelling stance toward technological exploration. Future work can build on this approach by studying AI-assisted composing practices and interaction patterns in different communities of practice and with different technologies and applications. Given the rapid pace of change, we can anticipate successive eras of transformation as AI capabilities and the interfaces for using them continue to evolve and integrate into learning and professional life.

Extensions of this approach could be especially valuable across writing studies contexts. How do threshold concepts for AI-assisted writing manifest: in first-year composition, where students learn foundational writing skills and academic discourse conventions; in professional and technical communication, where genre expectations and stakeholder relationships carry distinct stakes; and in writing across the disciplines, where faculty bring specialized expertise to guide AI integration within their fields? Each context will generate distinct tensions and insights. We encourage scholars across these communities to pursue ground-up investigations that allow threshold concepts to emerge from situated practice as a complement to applications of existing theory and frameworks. An ethics of experimentation commits us to continued inquiry with students, within disciplinary communities, and across the technological frontier where new thresholds will continue to emerge.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used the generative technologies, Anthropic's Claude and OpenAI's ChatGPT, in order to support ideation and to improve readability and language of the manuscript. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Abram D. Anders: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Emily Dux Speltz:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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